Cambridge Assessment International Education

Cambridge International AS & A Level

MATHEMATICS **9709/53**

Paper 5 Probability & Statistics 1 **May/June 2022**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathsematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (ISW).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

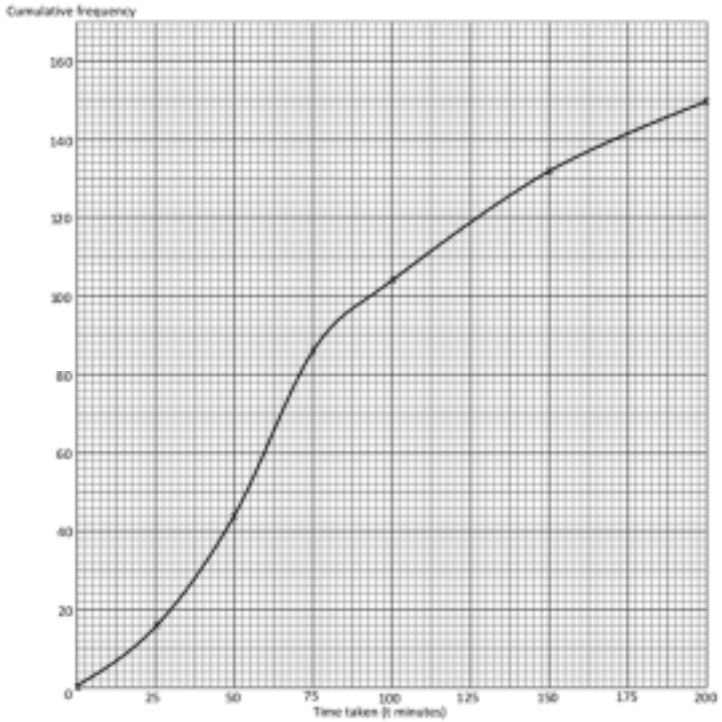
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1(a)	Cumulative frequency (cf) graph	**M1**	At least 3 points plotted accurately at class upper end points (25,16) (50,44) (75,86) (100,104) (150, 132) (200, 150). Linear cf scale $0 \leqslant cf \leqslant 150$ and linear time scale $0 \leqslant$ time(mins) $\leqslant 200$ with at least 3 values identified on each axis.
		**A1**	All points plotted correctly, curve drawn (within tolerance) and joined to (0,0). Axes labelled cumulative frequency (cf), time (t) and minutes (min), or a suitable title.
		2	
1(b)	Line from cumulative frequency = 30 to meet graph at t is between 37.5 and 42	**B1 FT**	Not from wrong working. Must be an increasing cumulative frequency graph.
		1	

Question	Answer	Marks	Guidance
2(a)	$\left[\frac{123.4}{20} = \right] 6.17$	**B1**	Accept 6 m 17 cm, $\frac{1234}{200}$.
		1	
2(b)	$\frac{10\text{th} + 11\text{th}}{2} = \frac{5.4 + 5.5}{2} = 5.45$ (m)	**B1**	Accept 5 m 45 cm.
		1	
2(c)	The mean is unduly influenced by an extreme value, 19.4.	**B1**	Comment must be within context.
		1	

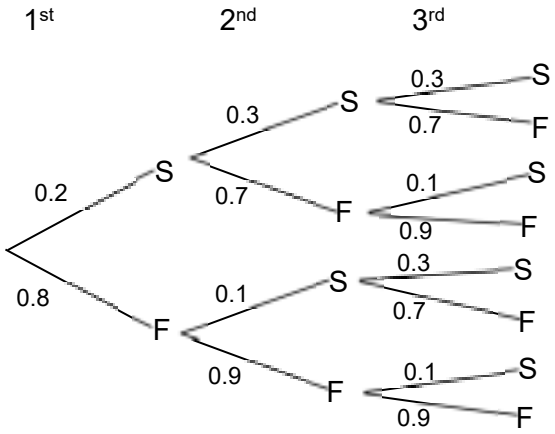
Question	Answer	Marks	Guidance
3(a)	$k = \frac{1}{18}$ $(4k + k + 4k + 9k = 18k = 1)$	**B1**	SOI
	<table><tr><td>x</td><td>–2</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X=x)</td><td>$\frac{4}{18}$</td><td>$\frac{1}{18}$</td><td>$\frac{4}{18}$</td><td>$\frac{9}{18}$</td></tr></table>	**M1**	Table with correct x values and at least one probability accurate using *their k*. Values need not be in order, lines may not be drawn, may be vertical, x and P(X=x) may be omitted. Condone any additional X values if probability stated as 0.
		A1	Remaining probabilities correct.
		3	

Question	Answer	Marks	Guidance
3(b)	$\left[\mathrm{E}(X)=\frac{4\times-2+1\times1+4\times2+9\times3}{18}=\right]$ $\frac{-8+1+8+27}{18}$	**M1**	$-8k+k+8k+27k$ May be implied by use in Variance. Accept unsimplified expression. FT *their* table if probabilities sum to 1 or 0.999. **SC B1** $28k$.
	$\left[\mathrm{Var}(X)=\frac{4\times(-2)^2+1\times1^2+4\times2^2+9\times3^2}{18}-\left(\textit{their}\,\mathrm{E}(X)\right)^2=\right]$ $=\frac{16+1+16+81}{18}-\left(\textit{their}\,\frac{28}{18}\right)^2$	**M1**	$16k+k+16k+81k-(\textit{their}\text{ mean})^2$ FT *their* table even if probabilities not summing to 1. Note: If table is correct, $\frac{114}{18}-\left(\textit{their}\,\mathrm{E}(X)\right)^2$ M1. **SC B1** $114k-(\textit{their}\text{ mean})^2$.
	$\mathrm{E}(X)=\frac{14}{9}, 1\frac{5}{9}, 1.56$, $\mathrm{Var}(X)=\frac{317}{81}, 3\frac{74}{81}, 3.91$	**A1**	Answers for E(X) and Var(X) must be identified. $3.91 \leqslant \mathrm{Var}(X) \leqslant 3.914$
		3	

Question	Answer	Marks	Guidance
4(a)	$\left[\left(\frac{5}{6}\right)^7\times\frac{1}{6}=\right]0.0465, \frac{78125}{1679616}$	**B1**	$0.0465 \leqslant p < 0.04652$
		1	

Question	Answer	Marks	Guidance
4(b)	$\mathrm{P}(X<6) = 1-\left(\frac{5}{6}\right)^5$ or $\frac{1}{6}+\left(\frac{5}{6}\right)\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^2\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^3\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^4\left(\frac{1}{6}\right)$	**M1**	$1-p^n$, $0<p<1$, $n = 4, 5, 6$ or sum of 4, 5 or 6 terms $p\times(1-p)^n$ for $n=0,1,2,3,4(5)$.
	0.598, $\frac{4651}{7776}$	**A1**	
		2	
4(c)	[Probability of total less than 4 is] $\frac{3}{36}$ or $\frac{1}{12}$	**B1**	SOI
	[1 – P(0, 1, 2)] $=1-\left({}^{10}\mathrm{C}_0\left(\frac{1}{12}\right)^0\left(\frac{11}{12}\right)^{10}+{}^{10}\mathrm{C}_1\left(\frac{1}{12}\right)^1\left(\frac{11}{12}\right)^9+{}^{10}\mathrm{C}_2\left(\frac{1}{12}\right)^2\left(\frac{11}{12}\right)^8\right)$	**M1**	One term ${}^{10}\mathrm{C}_x\ p^x(1-p)^{10-x}$, for $0<x<10$, $0<p<1$.
	1 – (0.418904 + 0.380822 + 0.155791)	**A1 FT**	Correct expression. Accept unsimplified.
	0.0445	**A1**	$0.04448 \leqslant p \leqslant 0.0445$
		4	

Question	Answer	Marks	Guidance
5(a)	$[\text{P}(142 < X < 205)] = \text{P}\left(\frac{142-170}{25} < z < \frac{205-170}{25}\right)$	**M1**	Use of ± standardisation formula once substituting 170, 25 and either 142 or 205 appropriately.. Condone 25^2 and continuity correction ±0.5.
	$\text{P}(-1.12 < z < 1.4)$	**A1**	Both correct. Accept unsimplified.
	$\Phi(1.4) - (1 - \Phi(1.12)) = 0.9192 + 0.8686 - 1$	**M1**	Calculating the appropriate area from stated phis of z-values.
	0.788	**A1**	AWRT, not from wrong working
		4	
5(b)	$\text{P}(X > 205) = 1 - 0.9192 = 0.0808$	**B1 FT**	Correct or FT from part **5(a).**
	$(0.0808 \times 0.30 + \textit{their}\ 0.788 \times 0.24) \times 20000$	**M1**	Correct or *their* 0.0808 × 0.30 × k + *their* 0.788 × 0.24 × k, k positive integer.
	[$]4266.24	**A1**	4265 < income ⩽ 4270, not from wrong working
		3	
5(c)	$[\text{P}(Z > \frac{w-182}{20}) = 0.72]$ $\frac{w-182}{20} = -0.583$	**B1**	$0.5828 \leqslant z \leqslant 0.583$ or $-0.583 \leqslant z \leqslant -0.5828$ seen.
		M1	182 and 20 substituted in ± standardisation formula, no continuity correction, not σ^2, $\sqrt{\sigma}$, equated to a z-value.
	$w = 170$	**A1**	$170 \leqslant w < 170.35$
		3	

Question	Answer	Marks	Guidance
6(a)		**B1**	First and second jumps correct with probabilities and outcomes identified.
		B1	Third jump correct with probabilities and outcomes identified.
		2	
6(b)	SFF $0.2\times0.7\times0.9 = 0.126$ FSF $0.8\times0.1\times0.7 = 0.056$ FFS $0.8\times0.9\times0.1 = 0.072$	**M1**	Two or three correct 3 factor probabilities added, correct or FT from part **6(a)**. Accept unsimplified.
	[Total = probability of 1 success =] 0.254 $\left(\frac{127}{500}\right)$	**A1**	Accept unsimplified.
	[Probability of at least 1 success = $1-0.8\times0.9\times0.9=$]0.352 $\left(\frac{44}{125}\right)$	**B1 FT**	Accept unsimplified.
	P(exactly 1 success | at least 1 success)= $\frac{their\,0.254}{their\,0.352}$	**M1**	Accept unsimplified.
	0.722, $\frac{127}{176}$	**A1**	$0.7215 < p \leqslant 0.722$
		5	

Question	Answer	Marks	Guidance
6(c)	$0.8\times0.9\times0.9\times0.1\times0.3\times0.3=0.005832$ [FFFSSS] $0.2\times0.3\times0.3\times0.7\times0.9\times0.9=0.010206$ [SSSFFF]	**M1**	$a \times b \times c \times d \times e \times f$ FT from *their* tree diagram. Either *a, b* and *c* all = 0.8 or 0.9 (at least one of each) and *d, e* and *f* all = 0.1 or 0.3 (at least one of each). Or *a, b, c* = 0.2 or 0.3 (at least one of each) and *d, e, f* = 0.7 or 0.9 (at least one of each).
		A1	Either correct. Accept unsimplified.
	[Total =] 0.0160[38]	**A1**	
		3	

Question	Answer	Marks	Guidance
7(a)	${}^{12}C_5 \times {}^{7}C_4$ [$\times {}^{3}C_3$]	**M1**	${}^{12}C_r \times q$, $r = 3, 4, 5$ q a positive integer > 1, no + or − .__
		M1	${}^{12}C_s \times {}^{12-s}C_t$ [$\times {}^{12-s-t}C_u$] $s = 3, 4, 5$; $t = 3, 4, 5 \neq s$; $u = 3, 4, 5 \neq s, t$
	Alternative method for question 7(a)		
	$\frac{12!}{5!\times3!\times4!}$	**M1**	12! ÷ by a product of three factorials.
		M1	$\frac{n!}{5!\times3!\times4!}$
	[792 × 35 =] 27 720	**A1**	CAO
		3	

Question	Answer	Marks	Guidance
7(b)	4! (Lizo) × 6! (Kenny) × 2! (Martin) × 2! (Nantes)	**M1**	Product involving at least 3 of 4!, 6!, 2!, 2!
	× 3! (orders of K, M and N)	**M1**	w × 3! , *w* integer > 1.
	414 720	**A1**	WWW CAO
		3	
7(c)	${}^{7}C_{4}$ (adults) × ${}^{4}C_{1}$ × ${}^{3}C_{1}$	**M1**	${}^{7}C_{4}$ × *b*, *b* integer > 1 no + or – .
	420	**A1**	
		2	
7(d)	K not L ${}^{5}C_{3}$ × ${}^{8}C_{3}$ = 560 L not K ${}^{5}C_{3}$ × ${}^{8}C_{3}$ = 560 L and K ${}^{5}C_{2}$ × ${}^{8}C_{3}$= 560	**M1**	${}^{8}C_{3}$(or ${}^{8}P_{3}$) × *c* for one of the products or ${}^{5}C_{3}$ (or ${}^{5}P_{3}$)× *c*, positive integer >1 for first 2 products only.
		M1	Add 2 or 3 correct scenarios only values, no additional incorrect scenarios, no repeated scenarios. Accept unsimplified.
	[Total or Difference=] 1680	**A1**	
	Alternative method for question 7(d)		
	Total no of ways – neither L nor K Total = ${}^{7}C_{4}$ × ${}^{8}C_{3}$ = 1960 Neither K nor L = ${}^{5}C_{4}$ × ${}^{8}C_{3}$ =280	**M1**	${}^{8}C_{3}$ × *c, c* a positive integer >1.
		M1	Subtracting the number of ways with neither from their total number of ways.
	[Total or Difference=] 1680	**A1**	

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Question	Answer	Marks	Guidance
7(d)	**Alternative method for question 7(d)**		
	Subtracting K and L from sum of K and L K $^6C_3 \times {}^8C_3 = 1120$ L $^6C_3 \times {}^8C_3 = 1120$ L and K $^5C_2 \times {}^8C_3 = 560$ 1120 + 1120 – 560 = 1680	**M1** **M1**	$^8C_3 \times c$, c a positive integer >1. Subtracting number of ways with both from sum of number of ways with K and number of ways with L.
	[Total or Difference=] 1680	**A1**	
		3	